Inductor Geometry With Improved Energy Density

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***Abstract*—The "constant-flux" concept is leveraged to achieve high magnetic-energy density, leading to inductor geometries with height significantly lower than that of conventional products. Techniques to shape the core and to distribute the winding turns to shape a desirable field profile are described for the two basic classes of magnetic geometries: those with the winding enclosed by the core and those with the core enclosed by the winding. A relatively constant flux distribution is advantageous not only from the density standpoint, but also from the thermal standpoint via the reduction of hot spots, and from the reliability standpoint via the suppression of flux crowding. In this journal paper on a constant-flux inductor (CFI) with enclosed winding, the foci are operating principle, dc analysis, and basic design procedure. Prototype cores and windings were routed from powder-iron disks and copper sheets, respectively. The design of CFI was validated by the assembled inductor prototype.**

***Index Terms*—Constant-flux, high energy density, inductor, low profile, uniform flux distribution.**

Nomenclature

α	Uniformity factor.
AT_j	Ampere-turns in winding window j.
$B_{\max}$	Maximum flux density.
E	Energy stored in the core.
H_c	Core thickness.
H_p	Plate thickness between winding and core.
H_w	Winding thickness.
I_r	Current rating.
L	Inductance of the inductor.
μ	Permeability of core material.
μ_0	Vacuum permeability.
n_j	Number of turns in winding window j.
N_w	Number of winding windows.
ρ_{Cu}	Copper resistivity.
R_c	Outer radius of the core.
$R_{\rm dc}$	DC resistance of the inductor.
R_{Oj}, R_{Ij}	Outer and inner radii of winding window j.

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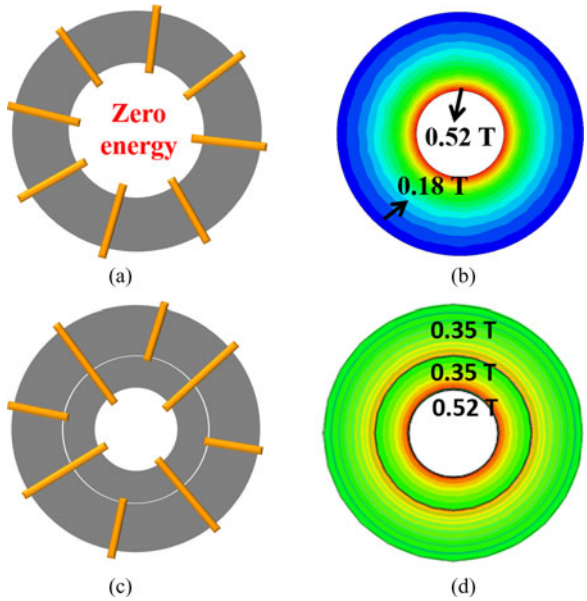


Fig. 1. (a) Conventional toroidal inductor with region storing no energy; (b) unevenly distributed flux density inside a toroidal core; (c) CFI with core enclosed by winding; and (d) more uniform flux density distribution demonstrating improved space utilization.

I. Introduction

RESEARCHERS in academia, government, and industry have been searching for ways to improve magnetic energy density and reduce package size of magnetic parts [1]–[12]. The matrix core, discussed in [2]–[4], consists of multiple cores with winding connected. This structure has low profile and good heat dissipation. However, it sees nonuniform flux density among elements. With the purpose of fully utilizing magnetic material and achieving a low profile inductor, the inductor structure proposed in [5] employs multiple permeabilities in order to achieve uniform flux density. Coupled inductor with a lateral flux structure revealed in [6] was proved to have higher density than vertical flux structure so that the inductor size can be further reduced. For toroidal inductors, adding concentric toroidal cells and threading the windings properly resulted in more uniform distribution of magnetic flux, significantly improving power density [8]. Fig. 1 shows that the constant-flux inductor (CFI) with core enclosed by winding has improved space utilization and more uniform flux distribution.

Compared to the inductor in [8], the CFI introduced herein has enclosed winding. The winding inside the core is configured to distribute the magnetic flux relatively uniformly throughout the magnetic volume to obtain higher energy density and smaller package volume.

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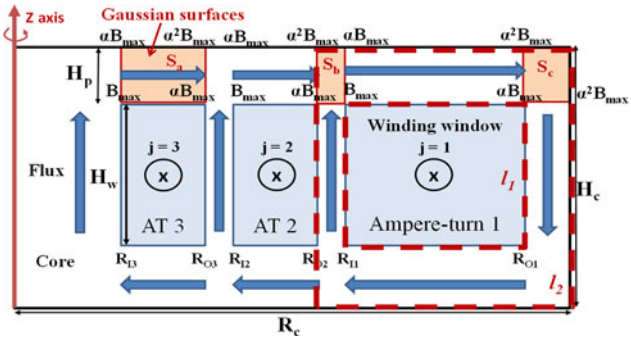


Fig. 2. Axisymmetric view of a CFI with enclosed winding showing three winding windows, ampere-turns direction, and magnetic flux path.

Section II describes the principle and mathematics for shaping constant flux, and derives the formulas for such key performance parameters as the inductance and dc resistance. Section III gives detailed design procedures to configure the structure of a CFI, as well as design examples to demonstrate the procedures. Section IV presents the experimental results of a fabricated CFI that validates the design example in Section III. While this paper is mainly about dc behavior, preliminary simulation results are reported in Section V for such practical issues such as the core loss and ac winding loss. Conclusions are also given in Section V.

II. Behavioral Model of a CFI

Based on the enclosed-winding geometry, the behavioral model of the CFI can be constructed in order to shape the flux distribution and estimate the performance of the inductor. As seen in Fig. 2, an ideal CFI with enclosed winding consists of a magnetic core with N_w winding windows carrying prescribed ampere-turns AT_j. The objective is to synthesize the ampere-turns to constrain the flux variation to a "uniformity factor α," ratio of the minimum flux density $\alpha B_{\max}$ to the maximum flux density $B_{\max}$ allowed in the magnetic field around winding window j. For a given core loss density and frequency of operation, the maximum magnetic flux density $B_{\max}$ can be determined from the material's magnetic property. "Constant" flux is achieved when α approaches unity. The CFIs in Fig. 3(a) and (c) were designed with the same parameters, but with different uniformity factor. It can be seen from Fig. 3(b) and (d) that the flux distribution is more uniform when α increases. This section delineates the basic laws and equations that are used to achieve a prescribed α, as well as the calculation of incremental inductance and dc resistance.

A. Geometrical Variables

The radii and thickness of the winding windows are considered as geometrical variables. These variables are solved through equations assuming that, as shown in Fig. 2, the flux density peaks at $B_{\max}$ at the inner radius of each winding window, and drops by a factor of α at the outer radius of the winding window. Between adjacent winding windows, the ampere-turns are added to inject flux to bring $\alpha B_{\max}$ to $B_{\max}$. The flux density is assumed to drop by a factor of α along the vertical direction of the magnetic region above the winding windows, and to be constant along the vertical edges of the winding windows. From Gauss' law, the amount of flux that flows into a surface equals to the amount of flux that flows out of the surface. Therefore, the behavioral model of the CFI can be derived based on the Gaussian surfaces drawn as shown in Fig. 2. The surface S_a encloses the magnetic volume from R_{Ij} to R_{Oj} above the winding windows; the surface S_b encloses the magnetic volume from $R_{O(j+1)}$ to R_{Ij} between two winding windows; and the surface S_c encloses the magnetic volume at the corners such as R_{O1} to R_c and 0 to R_{I3}. In the case of winding window $j = 1$, relationship among the geometrical variables is derived by integrating the flux density over S_c

$$2\pi R_{O1} H_p \frac{(\alpha+\alpha^2)B_{\max}}{2} = \pi(R_c^2 - R_{O1}^2)\frac{(\alpha+\alpha^2)B_{\max}}{2}$$

$$\text{or} \qquad R_{O1}^2 + 2R_{O1}H_p - R_c^2 = 0 \tag{1}$$

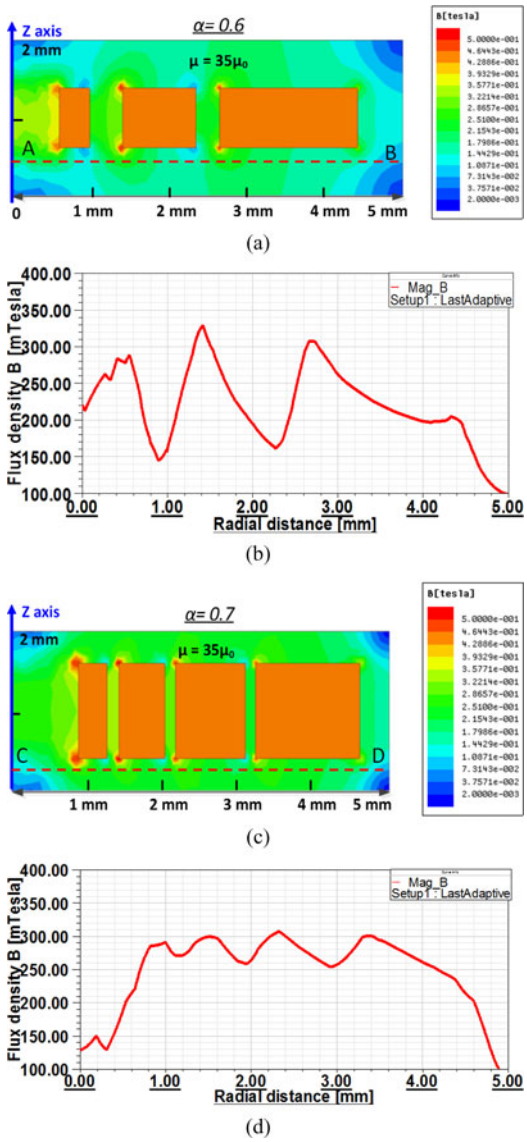


Fig. 3. Distributions [(b) and (d)] of magnetic flux density along lines AB in (a) with $\alpha = 0.6$ and CD in (c) with $\alpha = 0.7$ for a design with $R_c = 5$ mm, $H_c = 2$ mm, $B_{\max} = 0.35$ T, and $\mu = 35\mu_0$.

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where R_c is the outer radius of the core, which equals the footprint radius of the inductor and R_{O1} and H_p are the unknowns.

For the Gaussian surface S_a, the flux flows from the inner radius to the outer radius of winding window j. Therefore, for each winding window, the ratio of outer radius to the inner radius is related to the magnetic flux density by

$$2\pi R_{Oj} H_p \frac{\alpha+\alpha^2}{2} B_{\max} = 2\pi R_{Ij} H_p \frac{1+\alpha}{2} B_{\max}$$
$$\text{or } \frac{R_{Oj}}{R_{Ij}} = \frac{1}{\alpha}, \quad 1 \le j \le N_w. \tag{2}$$

For the Gaussian surface S_b, the equation can be written as

$$\frac{\alpha+\alpha^2}{2} B_{\max} 2\pi R_{O(j+1)} H_p + \frac{1+\alpha}{2} B_{\max} \pi (R_{Ij}^2 - R_{O(j+1)}^2) = \frac{1+\alpha}{2} B_{\max} 2\pi R_{Ij} H_p$$

or

$$R_{O(j+1)}^2 - 2\alpha R_{O(j+1)} H_p + 2R_{Ij} H_p - R_{Ij}^2 = 0 \qquad 1 \le j \le N_w. \tag{3}$$

Ampere's law is invoked to calculate H_p to ensure that magnetic flux density is reduced by α between the top surface of the core and the top surface of the winding. In the simulation, the flux lines follow a pattern and the corner areas are not necessarily filled with flux; however, in the behavioral model, the flux lines around the winding window are approximated to be rectangular as shown in Fig. 2 to simplify the calculation. Around closed contours l_1 and l_2

$$\int_{l_1} H \cdot dl_1 = \int_{l_2} \alpha H \cdot dl_2$$
$$H_w \cdot (1+\alpha) + 2 \cdot R_{I1} \cdot \ln\left(\frac{R_{O1}}{R_{I1}}\right) = \alpha \cdot (H_w + 2H_p)(1+\alpha) + 2 \cdot \alpha \cdot R_{O2} \cdot \ln\left(\frac{R_c}{R_{O2}}\right). \tag{4}$$

In the next section, (1)–(4) simultaneously solve the four variables R_{O1}, R_{I1}, R_{O2}, and H_p for the first winding window $j = 1$, and (2) and (3) recursively yield the radii of the remaining winding windows $j > 1$. Note that the following simplifications are included considering fabrication constraints.

1) Once H_p is determined from the outermost winding window, the value remains constant for the remaining winding windows. Ideally, each winding window j has a calculated H_p from (4), and the larger the j, the larger the H_p.
2) The inner radius of the innermost winding window is determined by (2), and should satisfy (3). Therefore, a determined amount of core material should be removed from the center. When the core center is left solid to simplify fabrication, error less than 3% was observed in the simulation.

For different uniformity factors from 0.5 to 0.8, the overall error in the inductance and resistance due to approximation and simplification in the design is less than 10% compared with the simulation result.

B. Number of Turns

To calculate the ampere-turns that should be applied to winding window j, closed contours like l_1 in Fig. 2 is drawn on every winding window. Based on Ampere's law, ampere-turn AT_j is assigned to winding window j by

$$AT_j = H_w \cdot \left(\frac{B_{\max}}{\mu} \pm \frac{\alpha B_{\max}}{\mu}\right) + 2 \cdot \frac{B_{\max}}{\mu} \cdot R_{Ij} \cdot \ln\left(\frac{R_{Oj}}{R_{Ij}}\right) \tag{5}$$

where H_w is the winding thickness given by $H_w = H_c - 2H_p$.

The number of turns in winding window j is determined by AT_j and the rated current I_r

$$n_j = \frac{AT_j}{I_r}. \tag{6}$$

Therefore, the total number of turns is the sum of the number of turns in each winding window

$$N = \sum_{j=1}^{N_w} n_j = \sum_{j=1}^{N_w} \frac{AT_j}{I_r} \tag{7}$$

where N_w is the total number of winding windows.

C. Incremental Inductance and DC Resistance

The incremental inductance and dc resistance can be calculated based on the ideal assumptions that the winding and the core are in circular shape and the effective flux density throughout the core is the average value of $B_{\max}$ and $\alpha B_{\max}$. Therefore, the energy stored in the CFI can be calculated by integrating the flux density throughout the total core volume

$$E \approx \frac{\pi B_{\max}^2}{2\mu} \cdot \left(\frac{1+\alpha}{2}\right)^2 \left(R_c^2 H_c - H_w \sum_{j=1}^{N_w} (R_{Oj}^2 - R_{Ij}^2)\right). \tag{8}$$

The inductance is related to the energy and current rating by

$$L = \frac{2E}{I_r^2}. \tag{9}$$

The current rating is related to the number of turns N and ampere-turns AT_j by (7), and the square of current rating can be expressed as follows after replacing AT_j by (5)

$$I_r^2 = \frac{1}{N^2} \left(\sum_{j=1}^{N_w} AT_j\right)^2 \tag{10}$$

$$I_r^2 = \frac{B_{\max}^2}{\mu^2 N^2} \left(\sum_{j=1}^{N_w} \left(H_w(1 \pm \alpha) + 2R_{Ij} \cdot \ln\left(\frac{R_{Oj}}{R_{Ij}}\right)\right)\right)^2. \tag{11}$$

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Therefore, by substituting E and I_r^2 by (8) and (11), the incremental inductance can be derived as

$$L = \frac{\mu N^2 \cdot \left(\frac{1+\alpha}{2}\right)^2 \cdot \pi (R_c^2 H_c - H_w \sum_{j=1}^{N_w} (R_{Oj}^2 - R_{Ij}^2))}{\left[\sum_{j=1}^{N_w} \left(H_w \cdot (1 \pm \alpha) + 2 \cdot R_{Ij} \cdot \ln\left(\frac{R_{Oj}}{R_{Ij}}\right)\right)\right]^2}. \tag{12}$$

If each winding turn were an ideal annulus with radii equal to the corresponding window radii, the dc resistance is given by [9]

$$R_{\rm dc} = \frac{\rho_{Cu} \cdot 2\pi}{H_w \ln(1/\alpha)} \cdot \sum_{j=1}^{N_w} n_j^2 \tag{13}$$

where ρ_{Cu} and H_w are the resistivity and thickness of the winding, respectively.

In practice, however, the winding window is split into the number of winding turns rounded from (6), and nonzero spacing among the turns has to be inserted horizontally and vertically to allow fabrication clearance. The footprint of the core could be square, and the winding turns could also be square to better utilize the material at the corners of the core (at the expense of resistance). The reduction in cross-sectional area of the winding owing to the clearances is modeled by segregating the clearances to the periphery of the winding window, and by lumping all conductive materials (e.g., copper) into a contiguous annulus with effective radii-radio α_j. The length of a square turn is assumed to be $4/\pi$ times than that of a circular turn. Thus, the effective dc resistance of a practical winding is obtained by modifying (13)

$$R_{\rm dc_e} \approx \frac{8\rho_{Cu}}{H_{w_e}} \cdot \sum_{j=1}^{N_w} \left(\frac{n_j^2}{\ln(1/\alpha_j)}\right). \tag{14}$$

D. Permeability and Current Rating

Since (1)–(4) are independent of the permeability of the core material, changing μ will not change the geometrical design. Therefore, the volume for energy storage remains the same. Equation (8) suggests that the total energy increases when the permeability decreases, resulting in an increase on inductance L for a given current rating I_r from (9). Fig. 4 gives an example of the relationships among current rating, inductance, and dc resistance when the relative permeability changes from 20 to 30. Along each curve, the current rating is fixed while the relative permeability is swept from 20 to 30. Either smaller permeability or smaller current rating would increase the number of turns that each winding window represents in order to maintain the maximum flux density $B_{\rm max}$. Therefore, the inductance and the resistance increase according to (12) and (13).

III. Design of a CFI

A. Design Procedure

The electrical inputs to the design include target inductance L_o, rated current I_r, and target dc resistance R_o. Practically, R_o is calculated from the loss budget and I_r. AC losses in the core and the winding, as well as temperature rise, are other practical issues that need to be addressed in the future.

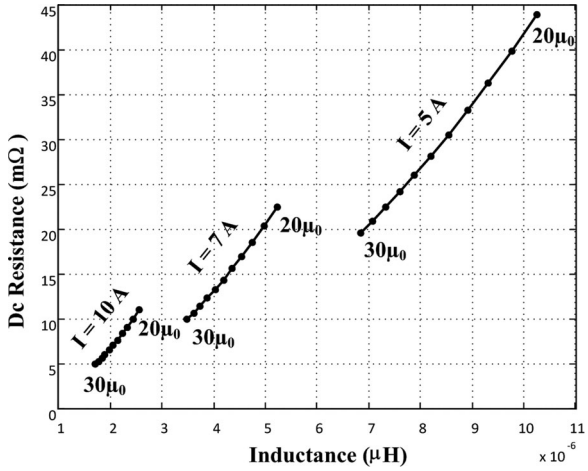


Fig. 4. Relationships among inductance, dc resistance, current rating, and permeability from $20\mu_0$ to $30\mu_0$ for $R_c = 5$ mm, $H_c = 2$ mm, $B_{\rm max} = 0.35$ T, and uniformity factor $\alpha = 0.6$.

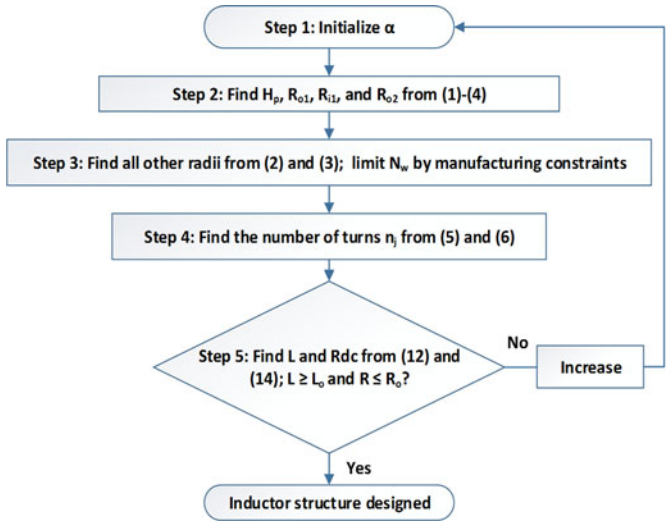


Fig. 5. Flowchart of the design procedure of a CFI.

The magnetic inputs include maximum magnetic flux density $B_{\rm max}$ and permeability μ. The mechanical inputs include the footprint radius R_c and inductor height H_c. The uniformity factor α is swept and all design variables are calculated until the required inductance L_o and dc resistance R_o are met. The procedure is described later with the flowchart in Fig. 5.

As the first step, the uniformity factor α is initialized at 0.5, which is the minimal value to maintain the benefit of constant flux. The plate thickness H_p, and radii R_{O1}, R_{I1}, and R_{O2} can be calculated by solving (1)–(4) simultaneously. After H_p is determined, the radii of the remaining winding windows can be obtained recursively from (2) and (3); note that the total number of winding windows N_w is limited by the manufacturing constraints such as the width of the winding and the distance between the winding and the center of the core. The number of turns n_j is approximated from (5) and (6). Since n_j is a rounded

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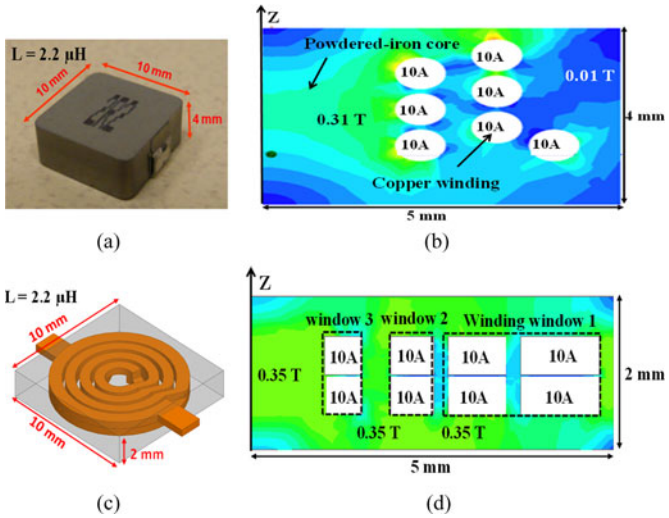


Fig. 6. (a) Commercial enclosed-winding inductor with inductance of 2.2 μH, resistance of 7 mΩ, height of 4 mm, and permeability of 35μ_0. (b) Axisymmetric view of the commercial inductor showing unevenly distributed flux density inside the core. (c) Three-dimensional structure of a CFI with winding enclosed by core; total thickness is reduced by half to 2 mm, whereas inductance and resistance are kept at 2.24 μH and 5.5 mΩ with 35μ_0 permeability. (d) Relatively uniform distribution of flux density in the core of the CFI.

integer, the approximation depends on the designer's preference and fabrication constraints. When n_j is rounded down, the flux uniformity would not be as uniform as the ideal design, and when n_j is rounded up, the flux density would be higher than the design, which may lead to saturation. Even number of turns is preferred since the winding turns can thus be divided into two layers. The thicknesses of the winding turns are the same, but the widths of the winding turns are not necessarily the same. As discussed in [9], the total dc resistance is minimized if the concentric winding turns has equal resistance, which means the outer winding turns have larger width than that of the inner turns. Finally, the inductance L and resistance $R_{\rm dc}$ can be found from (12) and (14). If the requirements cannot be met, increase the uniformity factor α and go back to the first step. If no α gives satisfied inductance and resistance, increase the dimension of the core R_c or H_c and repeat the procedure. If the calculated inductance and resistance are both better than the target, the inductor volume can be further reduced.

B. Design Examples

The CFI can be particularly advantageous in optimizing the space utilization of low profile inductors. In commercial enclosed-winding inductors, the average energy density is low since the flux density distribution throughout the volume is not constant. This section describes two examples of CFI design. The first one shows significant improvement in the utilization of magnetic material and size reduction compared to a commercial inductor, whereas the second example is used for fabrication and validation for the design of a CFI.

As a first example, Fig. 6(a) shows a commercial inductor rated at 2.2 μH, 7 mΩ, and 12 A [13]. The interior of this inductor was examined by cross sectioning so that the solid model in Fig. 6(b) could be constructed for finite-element analysis of the flux distribution throughout the core volume. After sweeping the permeability of the magnetic core, finite-element simulation suggested that the core material ought to have relative permeability of 35 to achieve the rated inductance. As shown in Fig. 6(b), the core volume is filled by large area with very low flux density, suggesting the magnetic material is not fully utilized. By following the design procedure in Section III-A, the inductor structure shown in Fig. 6(c) was designed and simulated using the same material properties and footprint dimensions as the structure simulated in Fig. 6(b). Inductor structure in Fig. 6(c) has the same footprint area, inductance, resistance, and current rating as the commercial inductor, but the total height is reduced by a factor of 2, thanks to flux uniformity as shown in Fig. 6(d).

TABLE I
DESIGN PARAMETERS OF A CFI

INPUTS						
B_{max} (T)	R_c (mm)	H_c (mm)	I_r (A)	L_{req} (μH)	R_{req} (mΩ)	μ (μ_0)
0.35	5	2	10	1.5	8	22
OUTPUTS (α = 0.65)						
Winding Window	R_{Oj} (mm)		R_{Ij} (mm)	AT_j (A)	n_j	
j = 1	4.5		2.9	52.6	4	
j = 2	2.7		1.7	23.7	2	
j = 3	1.5		1.0	15.3	2	
Total thickness of winding (H_w)	1 mm					
Inductance calculated (L)	1.6 μH					
Dc resistance calculated (R_{dc})	7.0 mΩ					

The second example was designed and used for fabrication as described in Section IV in order to validate the concept. The core material of Micrometals no. 8 was used for fabrication, and a toroid was made from the plain core in order to measure the permeability. The relative permeability was measured by using the impedance analyzer Agilent 4294A, terminal adapter Agilent 42942A, test fixture Agilent 16454A, and the measurement result was 22. Therefore, $22\mu_0$ was used in the design, and the maximum flux density $B_{\rm max}$ was specified as 0.35 T to keep the core away from saturation. The target inductance was 1.5 μH, and the target resistance was 8 mΩ. The procedure outlined earlier yielded an optimized design with $L = 1.6$ μH, $I_r = 10$ A, and $R_{\rm dc} = 7.0$ mΩ in a volume of $10 \times 10 \times 2$ mm^3.

Following the design procedure in Section III-A for $\alpha = 0.65$, after solving (1)–(4), the plate thickness H_p was 0.5 mm, the outer radius of window $j = 1$ was 4.5 mm, and the inner radius was 2.9 mm. Solving (2) and (3) for the next windows, the radii can be determined successively, and then the ampere-turns in each window AT_j can be calculated from (5) which were 52.6 A, 23.7 A, and 15.3 A. The numbers of turns in each window n_j were approximated to 4, 2, and 2 from the 10 A current rating I_r using (6), so that they can be distributed into two layers. Table I summarizes all the final design results.

C. Three-Dimensional Finite-Element Verification

The parameters listed in Table I were used to construct the ideal structure of the CFI, as shown in Fig. 7(a). The simulation

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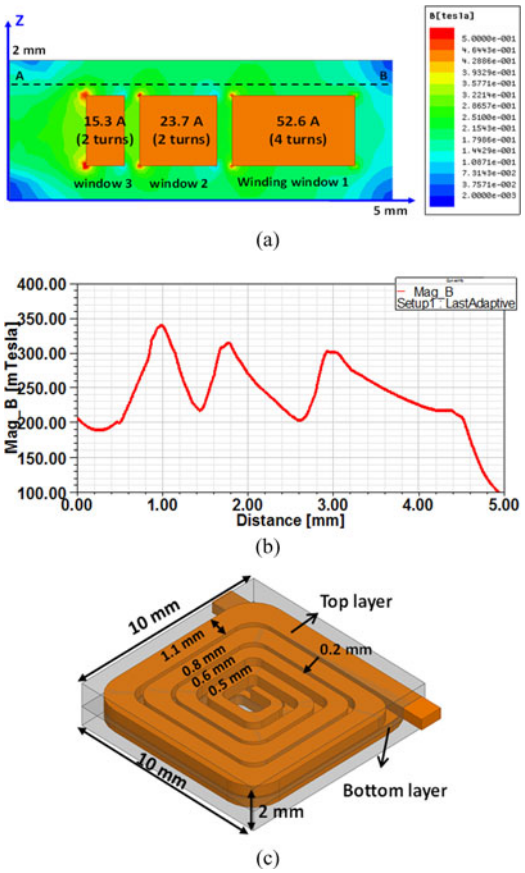


Fig. 7. (a) Relatively uniform flux density distribution ($\alpha = 0.65$) across an axisymmetric cross section using the designed parameters in Table I. (b) Distribution of flux density along line AB. (c) Solid model of the CFI in Maxwell with square-shape winding; 0.2 mm clearance was inserted between the winding turns in each layer due to fabrication limitations.

result of flux distribution shows no significant flux crowding as the peak magnetic flux density was limited to 0.35 T. The flux density along line AB is shown in Fig. 7(b), and the uniformity of the flux distribution was consistent with $\alpha = 0.65$. Based on the ideal structure, a solid model was constructed for fabrication as illustrated in Fig. 7(c). For fabrication considerations, the winding was configured to have two layers which started from the top terminal, spiraled counterclockwise toward the center, connected to the bottom winding layer at the center, and spiraled counterclockwise to the bottom terminal. The spacing between adjacent winding turns was filled with magnetic materials and was kept at not less than 0.2 mm due to limitation of a prototyping equipment. The outermost winding window was split into two turns per layer, and the width of the outermost turn was 0.3 mm larger than the inner turn so that the total resistance was minimized as discussed in Section III-A.

IV. Experimental Validation

A. Core Fabrication and Winding Fabrication

Micrometal's no. 8 plain core made of iron powder was used to fabricate the inductor example in Section III. As shown in Fig. 8(a), a cylinder plain core was sliced into 1-mm-thick disks

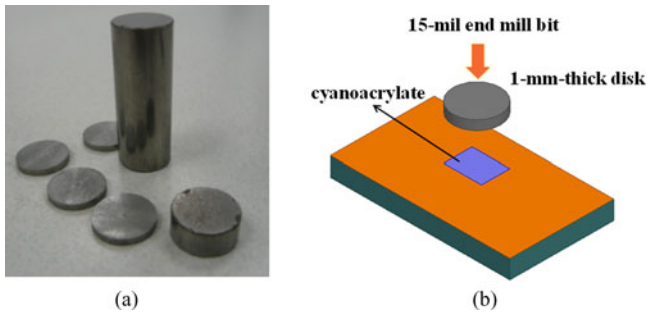


Fig. 8. (a) Plain core and core disks. (b) Preparation for the routing process.

in order to fabricate a core half. The core disk [see Fig. 8(b)] was bonded to a printed-circuit board using cyanoacrylate to stabilize the core during the routing process. The winding groove in each core half was routed by a T-Tech QC-5000 prototyping machine using a 15-mil square end-mill bit. The empirical settings of the prototyping machine for 1-mm-thick core disk are a spindle speed of 15 000 rounds/min and a routing velocity of 2 in/min. After routing, the core was released by soaking in acetone for half an hour. The methodology proposed here is for prototyping purpose, and an advanced manufacturing method need to be explored for mass production.

Copper sheets with 0.5 mm thickness were used to fabricate the winding by following the similar routing process as described in Fig. 8. However, before the copper was bonded to the printed-circuit board for routing, the copper sheet was pressed by Carver 2112 laminating press under 6000 lbf/in^2 for 15 min in order to ensure the flatness of the copper. The empirical settings of the prototyping machine for 0.5-mm-thick copper sheet were the same as that for the core, but the routing depth was kept at 0.05 mm each time to prevent the mill bit from breaking. After ten times of routing, the patterned copper winding was released from the substrate by soaking in the acetone for half an hour. Fig. 9(a) shows the prototypes of the core and the copper winding made by the routing method.

B. Inductor Assembly

The inductor half shown in Fig. 9(b) was obtained after the winding half was inserted into the corresponding core half. Two inductor halves were connected in the center by solder. A small amount of solder paste SMD291SNL10 was deposited in the center of the winding halves. The two inductor halves were placed onto a reflow belt furnace to carry out the solder process described in [14]. In order to exclude the air between the two halves as much as possible, an equivalent pressure of 500 lbf/in^2 was placed on top of the inductor. The prototype of the assembled inductor is shown in Fig. 9(c).

The concept of a CFI was validated by the measurement result on the prototype. The measured inductance and dc resistance of the fabricated CFI were 1.42 μH and 8 mΩ, respectively. Due to the limitation of the fabrication process, an inevitable air gap between the two inductor halves would be introduced since the solder in the center cannot be ideally flat. Therefore, the measured inductance (1.42 μH) was 12% smaller than the

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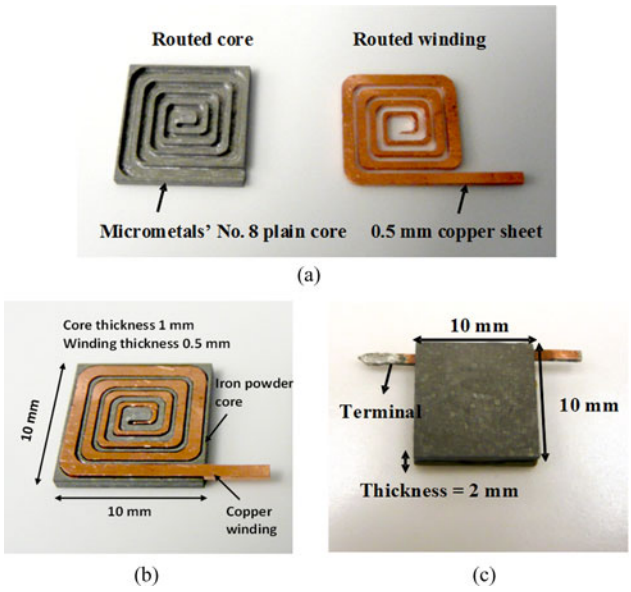


Fig. 9. Prototypes of (a) routed core and winding, (b) inductor half, and (c) assembled CFI.

simulation result (1.6 μH). The copper winding was trimmed to be slimmer than the simulated structure in order to fit into the core half, and the resistance measured (8 mΩ) was 14% larger than the simulated result. The performance would be closer to the design if the fabrication method can be improved.

V. Conclusion and Future Work

This paper describes a method to distribute the magnetic flux uniformly, improving energy density of inductors. A recursive design procedure has been delineated to realize a physical structure meeting specified inductance, dc resistance, and peak magnetic flux density. A design example demonstrates that the inductor volume can be reduced to 50% with the same inductance, resistance, current rating, and core material. The concept was validated by a CFI prototype with a measurement result of 1.4-μH inductance and 8-mΩ dc resistance.

The ac performance will be evaluated in the future, and some preliminary results are reported herein. In order to compare the ac winding loss and core loss of the CFI with the commercial product, the structures shown in Fig. 6(a) and (c) were simulated under 500 kHz frequency and 1-A-peak ac current with the same properties of the material. Preliminary simulation results are shown in Table II. While the volume of the CFI was two times smaller than that of the commercial inductor, the simulated core loss of the CFI was 40% smaller. The ac flux distribution was similar to that of the dc simulation shown in Fig. 6(b) and (d); the CFI had a more evenly distributed ac flux density than the commercial product. As for the temperature, the CFI had less area to dissipate the heat compared with the commercial inductor with larger volume, but core loss at the interior of the core and winding loss would have a shorter distance to travel to the surface thanks to the thinner profile. The heat flux was calculated by dividing the core loss over the total surface area, and the result of CFI was 20% smaller than that of the commercial. Both dc and ac winding resistances were lower for the CFI. Whether the CFI has an advantage on the heat dissipation and the larger signal performance is to be studied in the future work.

TABLE II
Comparisons of AC Simulation Results

	Commercial inductor [13]	Constant-flux inductor
Volume (mm^3)	10 × 10 × 4	10 × 10 × 2
Core loss (W)	0.56	0.35
Heat flux (mW/mm^2)	1.55	1.24
Dc winding resistance (mΩ)	7.0	5.5
Ac winding resistance (mΩ)	36	30

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